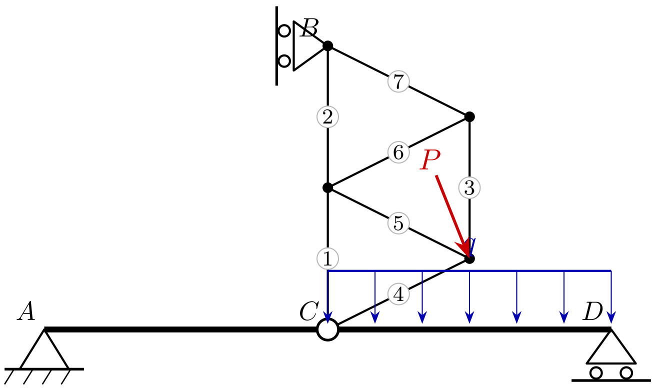
University of Natural Resources and Life Sciences, Vienna (BOKU)

VU Mechanik – LAWI 100515 (PI)

Exam **Group A** Date: 26.06.2026

Model solution



1. Static determinacy

Counting criterion $n = 3k - (a + z)$ (bodies k, reactions a, interaction forces z):

$$n = 3k - (a + z) = 3 \cdot 2 - (4 + 2) = 0 \;\Rightarrow\; \text{statically determinate}$$

2. Support reactions and hinge forces

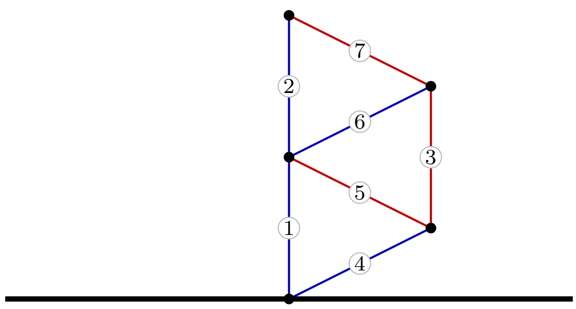
$A: \; R_x = 4, \; R_y = 13 \quad D: \; R_x = 0, \; R_y = 19 \quad B: \; R_x = -12, \; R_y = 0$

$C: \; H_x = -4, \; H_y = -20$

3. Truss member forces

Member	Force S [kN]	Type
1	-18	compression
2	-6	compression
3	12	tension
4	-4.47	compression
5	13.42	tension
6	-13.42	compression
7	13.42	tension

red: tension (+) blue: compression (−)



4. Section-force diagrams of the beams